

YEAR 10 SCIENCE (ADJUSTED)
PHYSICAL SCIENCES
REVISION - SEM. 2 EXAM 2019

1. A F1 car accelerates from rest at the Singapore Grand Prix and reaches the first corner 350 m away in 6.0 s with a final speed of 73 m/s. Calculate the acceleration of the vehicle.

(Use $a = \frac{v-u}{t}$)

$$v = 73 \text{ m/s}$$

$$u = 0 \text{ m/s}$$

$$a = ?$$

$$t = 6.0 \text{ s}$$

$$\begin{aligned} a &= \frac{v-u}{t} \\ &= \frac{73-0}{6.0} \\ &= \underline{12.2 \text{ m/s}^2} \end{aligned}$$

2. A person is taking photos from a balloon hovering 300 m above a paddock near Northam. She accidentally drops her phone and watches it smash into the ground 7.8 s later. Assume the initial velocity of the speed is zero and the acceleration due to gravity is 9.8 m/s^2 . How fast was the phone moving when it hit the ground? (Use $v = u + at$)

$$v = ?$$

$$u = 0 \text{ m/s}$$

$$a = 9.8 \text{ m/s}^2$$

$$t = 7.8 \text{ s}$$

$$\begin{aligned} v &= u + at \\ &= 0 + (9.8)(7.8) \\ &= \underline{76.4 \text{ m/s forwards}} \end{aligned}$$

3. A dragster moving at 92 m/s at the end of a race pops its parachute and uses its brakes to come to a stop in 6.0 s.

(a) What is its deceleration? (Use $a = \frac{v-u}{t}$)

$$v = 0 \text{ m/s}$$

$$u = 92 \text{ m/s}$$

$$a = ?$$

$$t = 6.0 \text{ s}$$

$$\begin{aligned} a &= \frac{v-u}{t} \\ &= \frac{0-92}{6.0} \\ &= \underline{-15.3 \text{ m/s}^2 \text{ backwards}} \end{aligned}$$

- (b) If the dragster has a mass of 930 kg, what force was applied by the parachute and the brakes?

$$F = ?$$

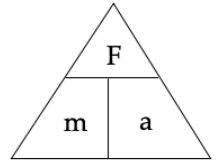
$$m = 930 \text{ kg}$$

$$a = -15.3 \text{ m/s}^2$$

$$F = ma$$

$$= (930)(-15.3)$$

$$= \underline{-14,230 \text{ N backwards}}$$



4. Complete these statements about Newton's three laws of motion.

- (a) An object remains at rest, or will not change its speed or direction, unless ...

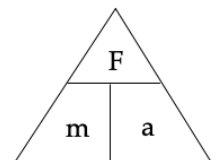
acted on by an external force.

- (b) When an unbalanced force acts on an object, the mass of an object affects ...

its acceleration.

- (c) For every action, there is ...

an equal and opposite reaction.



5. Complete the following table.

Force	Mass	Acceleration
400 N	<i>133 kg</i>	3 m/s ²
<i>42 N</i>	6.0 kg	7 m/s ²
4.8 N	0.9 kg	<i>5.3 m/s²</i>

6. (a) What is *inertia*?

An object at rest wants to stay still. A object that is moving wants to keep moving.

- (b) Which has the most inertia - a truck or a car?

truck - it has the bigger mass.

- (c) Which is the easier to stop quickly with its brakes? Explain why.

- Car*
- Smaller mass so the force applied can decelerate it quickly.*

7. (a) What happens to you when you are standing on the train and it suddenly takes off from the station?

- you move backwards.*

- (b) Why does this happen?

- your inertia wants to keep you stationary while the train moves forwards under you.*

8. Use Newton's Second Law to calculate answers for the following.

- (a) A 1700 kg car accelerates at 4.9 m/s^2 . What size force is needed to cause this acceleration?

$$F = ?$$

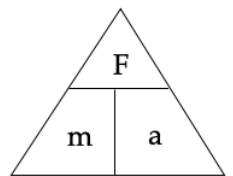
$$m = 1700 \text{ kg}$$

$$a = 4.9 \text{ m/s}^2$$

$$F = ma$$

$$= (1700)(4.9)$$

$$= \underline{8330 \text{ N forwards}}$$



- (b) A force of 250 N causes an object to accelerate at 3.0 m/s^2 . What is the object's mass?

$$\begin{aligned} F &= 250 \text{ N} \\ m &= ? \\ a &= 3.0 \text{ m/s}^2 \end{aligned} \quad \begin{aligned} F &= ma \\ \Rightarrow m &= \frac{F}{a} \\ &= \frac{250}{3.0} \\ &= \underline{83.3 \text{ kg}} \end{aligned}$$

- (c) A force of 450 N acts on a mass of 150 kg. What is the acceleration?

$$\begin{aligned} F &= 450 \text{ N} \\ m &= 150 \text{ kg} \\ a &= ? \end{aligned} \quad \begin{aligned} F &= ma \\ \Rightarrow a &= \frac{F}{m} \\ &= \frac{450}{150} \\ &= \underline{3.0 \text{ m/s}^2 \text{ forwards}} \end{aligned}$$

9. (a) Explain the difference between **mass** and **weight**.

MASS: the amount of matter in an object.
WEIGHT: force due to gravity on an object.

- (b) Are you lighter or heavier on the Moon? Explain why.

- lighter
- less gravity on the Moon.

- (c) The acceleration due to gravity on Jupiter is 22.5 m/s^2 . Calculate the weight of a 450 kg object on Jupiter.

$$F_w = ?$$

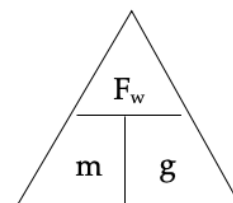
$$m = 450 \text{ kg}$$

$$g = 22.5 \text{ m/s}^2$$

$$F_w = mg$$

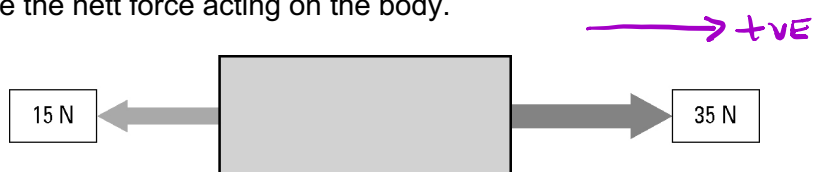
$$= (450)(22.5)$$

$$= \underline{10125 \text{ N}}$$



10. Consider the forces acting on a body of mass 20 kg resting on a frictionless surface.

- (a) Calculate the net force acting on the body.



$$\Sigma F = 35 - 15$$

$$= \underline{20 \text{ N to the right}}$$

- (b) Calculate the acceleration of the body.

$$F = 20 \text{ N}$$

$$m = 20 \text{ kg}$$

$$a = ?$$

$$F = ma$$

$$\Rightarrow a = \frac{F}{m}$$

$$= \frac{20}{20}$$

$$= \underline{1.0 \text{ m/s}^2 \text{ to the right}}$$

